# Ultimate Car Controller Guide

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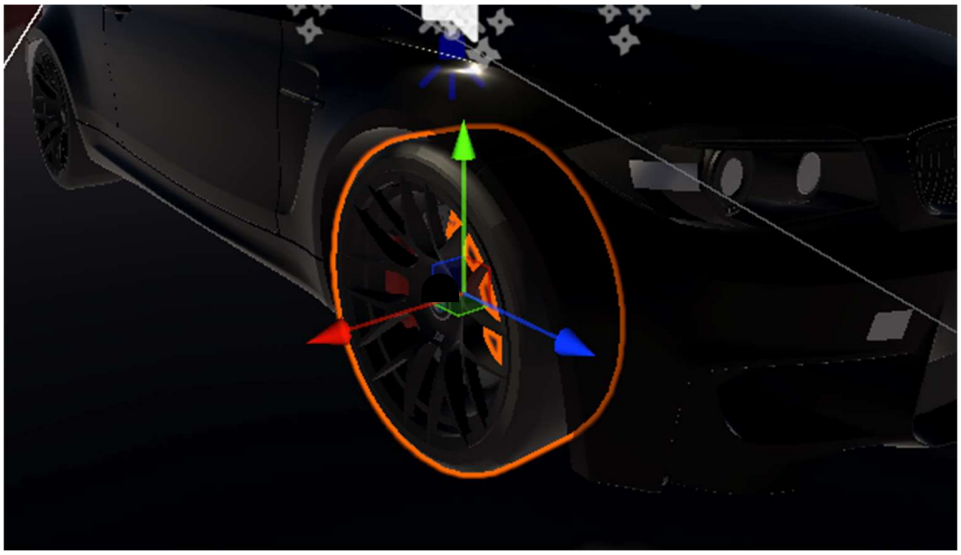
# 1.Using Car Editor

You can find the car editor at Tools/CarEditor.

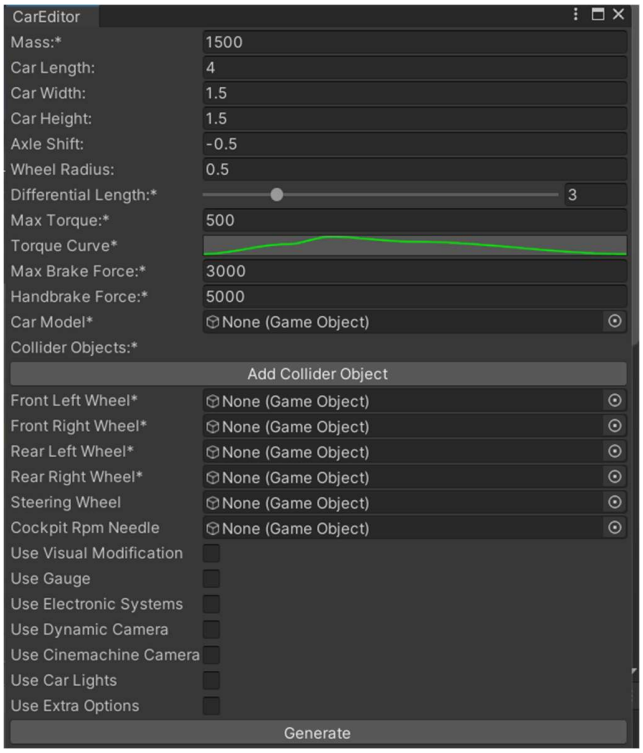
**If you are using prefab unpack completely.**

It is created to use specific presets to help you quickly create your vehicle and to adapt the parameters that come from you correctly**.**

**The forward side of the vehicle and wheels must face the direction the vehicle is going to go.**



Drag your car model to the scene. Make sure that there is no UCC script in the game object that is created. Fill in all required fields by dragging and dropping from the model. Click the Generate button.




You can quickly create your vehicle in under 30 seconds by filling in the required parameters.

## 1.1) Car Editor Parameters

1. Mass (Required)

Description: The weight of your vehicle.

Details: This value is assigned to the Rigidbody component of the main object. A realistic mass ensures the vehicle behaves naturally under physics simulation.

Recommendation: Use real-world vehicle mass values, e.g., 1500 kg for a typical car.

2. Car Length

Description: The longitudinal length of the car.

Details: Determines the size of the vehicle model in the forward and backward direction. Has an impact on performance.

Recommendation: Set based on the real-world length of your car model (e.g., 4 meters for a sedan).

3. Car Width

Description: The lateral width of the car.

Details: Defines how wide the car model is. Has an impact on performance.

Recommendation: Use real-world values for accuracy (e.g., 1.5 meters).

4. Car Height

Description: The vertical height of the car.

Details: Determines the height of the car model for collision and physics calculations. Has an impact on performance.

Recommendation: Typical cars have a height of around 1.4-1.5 meters.

5. Axle Shift

Description: Adjusts the position of the car's axles.

Details: Shifting the axle forward or backward can align the wheels with the car's visual model.

Recommendation: Fine-tune this value for precise wheel placement.

6. Wheel Radius

Description: The radius of the car wheels.

Details: Impacts wheel size and its interaction with the ground.

Recommendation: Use realistic values (e.g., 0.5 meters for average-sized wheels).

7. Differential Length(Required)

Description: Adjusts the length of the differential.

Details: This impacts the spacing between wheels.

Recommendation: Set based on your car's design; 3 units is a common default.

8. Max Torque(Required)

Description: The maximum rotational force applied to the wheels.

Details: Higher torque enables faster acceleration.

Recommendation: Set based on the car type (e.g., 300 Nm for typical cars, 500+ Nm for high-performance vehicles).

9. Torque Curve(Required)

Description: A graph representing how torque changes with RPM.

Details: Defines the engine's power output at various RPM levels.

Recommendation: Customize for your car type; a gradual curve is common for most vehicles.

10. Max Brake Force(Required)

Description: The maximum braking force applied by the vehicle.

Details: Determines how quickly the car can stop.

Recommendation: Use a realistic range, such as 3000 N for standard cars.

11. Handbrake Force(Required)

Description: The force applied when using the handbrake.

Details: Higher values allow for sharper, more controlled skids.

Recommendation: Adjust based on gameplay needs (e.g., 5000 N for dramatic drifts).

12. Car Model(Required)

Description: The 3D model of the car.

Details: Assign the visual representation of your vehicle.

Recommendation: Use optimized models for better performance.

13. Collider Objects(Required)

Description: Additional collider objects for the car.

Details: Helps improve collision detection and physics accuracy. Car system won’t work without at least one collider attached.

Recommendation: Assign colliders to match the car's shape. Objects such as chassis and bumper must be assigned.

14. Front Left Wheel(Required)

Description: Assign the GameObject representing the front-left wheel and adds Wheel Collider on it.

15. Front Right Wheel(Required)

Description: Assign the GameObject representing the front-left wheel and adds Wheel Collider on it.

16. Rear Left Wheel(Required)

Description: Assign the GameObject representing the front-left wheel and adds Wheel Collider on it.

17. Rear Right Wheel(Required)

Description: Assign the GameObject representing the front-left wheel and adds Wheel Collider on it.

18. Steering Wheel

Description: Assign the GameObject representing the car's steering wheel.

Details: Allows the steering wheel to rotate based on player input.

19. Cockpit RPM Needle

Description: Assign the needle GameObject for RPM visualization.

Details: Animates based on engine RPM to simulate a tachometer.

Recommendation: UCC_Gauge must be active. Rotation values use the rotation values given to the UI screen by UCC_Gauge

20. Use Visual Modification

Description: Toggle for enabling visual customization.

Details: Activates settings Camber, toe, spacer, smoke color.

21. Use Gauge

Description: Toggle for enabling a gauge system.

Details: Displays speed, RPM, or other stats in the HUD.

22. Use Electronic Systems

Description: Toggle for enabling electronic controls.

Details: Adds systems like ABS or traction control.

23. Use Dynamic Camera

Description: Toggle for enabling dynamic camera behavior for standard Unity Camera.

Details: Activates camera motion during acceleration, braking, or drifting.

Recommendation: Enable only if you don’t want to use Cinemachine.

24. Use Cinemachine Camera

Description: Toggle for enabling Cinemachine cameras.

Details: Allows advanced camera transitions and tracking.

Recommendation: Suggested camera system for UCC.

25. Use Car Lights

Description: Toggle for enabling lights like, headlights and brake lights.

Details: Activates light effects based on player inputs.

26. Use Extra Options

Description: Toggle for enabling additional features.

Details: Adds extras like exhaust fumes.

27. Generate

Description: Button to generate the car configuration.

Details: Creates and applies the car setup based on the provided parameters.

Recommendation: Ensure all fields are set correctly before clicking.

## 2. Scripts explained

### 2.1. UCC_CarController

It is the basic particle of the car system. It is automatically assigned by the editor. If the editor is not used, it must be added to the main object. Wheel colliders, moving parts, and communicating scripts are assigned here.

### 2.2. UCC_CarSettings

It is a script where all settings related to the car are made. All settings from car setup to suspension settings and torque settings are here. Its use is mandatory.

### 2.3. UCC_Calipers

Allows the calipers to follow the wheel movements.

### 2.4. UCC_CarLights

Provides control of 2 brakes, 2 forward, 2 reverse, an interior and an underglow light.

2.5. UCC_CinemachineCamera

This is the script that controls the Cinemachine cameras. Must be assigned to the main camera.

2.6. UCC_DynamicCamera

Standard camera controls for systems that do not use a Cinemachine camera. Must be assigned to the main camera.

2.7. UCC_ElectronicSystems

It is the system required for the use and settings of systems such as ABS, ESC, TC.

2.8. UCC_EngineSounds

It is my system that simulates sounds coming from the car. If created with the editor, all AudioSources are created automatically. Clips must be assigned manually.

2.9. UCC_ExtraCarOptions

This is the script that controls extra settings such as exhaust smoke.

2.10. UCC_Gauge

Script that controls the indicators in the cockpit and on the UI screen

2.11. UCC_VisualModification

Script that controls settings such as camber, toe, spacer, wheel smoke color.

2.12. UCC_CollisionHandler

This is the script required for visual deformation. It should be in the same object as the Rigidbody. If you are using an editor, it will assign your collider objects. You can also assign manually. You can adjust the damage intensity with this script.

3. Input System

3.1) Key Binding

Desired key assignments can be made from the Ultimate Car Controller/ Input System/ UCC_InputSystem file.

3.2) Gampad Support

All default features are supported by the gamepad. A separate assignment must be made for adding extra buttons.

3.3. Mobile Input System

Brake, gas, direction keys, n2o, handbrake are supported by the mobile system. Thanks to its flexible structure, new input can be added easily by creating a button.

**It is recommended to use the editor.**

**If there is no special reason, delete everything that may be related to the system, including the car model and all cameras, and start over.**

**It is recommended to use the Cinemachine camera system.**

**If you are not going to use the editor, make sure the necessary fields are assigned.**

For all your questions and comments, you can send an e-mail to
support@kairadigitalarts.com

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